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A Novel Unified Lightweight Temporal-Spatial Transformer Approach for Intrusion Detection in Drone Networks

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Abstract

The growing integration of drones across commercial, industrial, and civilian domains has introduced significant cybersecurity challenges, particularly due to the susceptibility of drone networks to a wide range of cyberattacks. Existing intrusion detection mechanisms often lack the adaptability, efficiency, and generalizability needed for the dynamic and resource-constrained environments in which drones operate. This paper proposes TSLT-Net, a novel lightweight and unified Temporal-Spatial Transformer-based intrusion detection system tailored specifically for drone networks. By leveraging self-attention mechanisms, TSLT-Net effectively models both temporal patterns and spatial dependencies in network traffic, enabling accurate detection of diverse intrusion types. The framework includes a streamlined preprocessing pipeline and supports both multiclass attack classification and binary anomaly detection within a single architecture. Extensive experiments conducted on the ISOT Drone Anomaly Detection Dataset, consisting of over 2.3 million labeled records, demonstrate TSLT-Net's superior performance with 99.99% accuracy in multiclass detection and 100% in binary anomaly detection, all while maintaining a minimal memory footprint of just 0.04 MB and 9,722 trainable parameters. These results establish TSLT-Net as an effective and scalable solution for real-time drone cybersecurity, particularly suitable for deployment on edge devices in mission-critical UAV systems.

Keywords: TSLT-Net; Drone intrusion detection; Temporal-spatial transformer; Lightweight deep learning; Transformer-based intrusion detection

1. Introduction

The widespread deployment of drones across sectors such as transportation, surveillance, agriculture, and logistics has significantly expanded the attack surface for cyber threats targeting unmanned aerial systems. As drones increasingly rely on wireless networks for communication and coordination, they become highly vulnerable to a range of cyberattacks, including denial-of-service (DoS), man-in-the-middle (MITM) attacks, spoofing, payload manipulation, and replay attacks [1], [2]. These intrusions can severely compromise operational safety, data integrity, and mission success, posing risks not only to the drones themselves but also to surrounding infrastructure and human populations [3], [4]. Despite the growing recognition of these

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threats, current security mechanisms are inadequate for reliably detecting and mitigating drone-specific intrusions, leaving critical gaps in the protection of drone networks [5], [6].

Addressing the security vulnerabilities of drone networks is essential to ensure the safe and reliable integration of drones into critical operations and public environments. Without effective intrusion detection, malicious actors can exploit network weaknesses to disrupt missions, steal sensitive data, or cause physical harm, undermining trust in drone technologies and limiting their broader adoption [7], [8]. Developing robust and efficient detection mechanisms will not only enhance the resilience of drone systems but also enable industries and governments to deploy drones with greater confidence, unlocking the full potential of drone-based services while safeguarding public safety and infrastructure [9], [10].

Detecting intrusions in drone networks presents unique challenges due to the dynamic, resource-constrained, and heterogeneous nature of these systems. Unlike conventional networks, drones operate under strict latency, bandwidth, and energy constraints, limiting the feasibility of computationally intensive security solutions [11], [12]. Additionally, the diversity of attack vectors, rapidly evolving threat landscapes, and the scarcity of large, labeled datasets tailored to drone environments make it difficult to train accurate and generalizable detection models. Furthermore, the need to distinguish subtle anomalies from normal drone behavior demands advanced methods capable of capturing both temporal and spatial dependencies within network traffic, which many traditional and deep learning approaches fail to address effectively [13].

The increasing adoption of drones in diverse sectors, such as logistics, surveillance, and emergency response, has significantly expanded the attack surface in drone communication networks. These systems are highly constrained in terms of latency, computational capacity, and energy, which limits the feasibility of traditional, resource-intensive security solutions. At the same time, the variety and sophistication of cyber threats targeting UAVs continue to evolve, necessitating intrusion detection systems that are not only accurate but also lightweight and scalable for real-time deployment. To address these challenges, this study introduces TSLT-Net, a novel intrusion detection framework based on a lightweight Temporal-Spatial Transformer. The main contributions of this work are summarized as follows:

- This is the first time we introduce a novel lightweight Transformer architecture that models temporal and spatial patterns in drone network traffic for intrusion detection.
- We incorporate an innovative reshaping and multi-head self-attention mechanism that captures sequential dependencies in non-sequential network traffic data, a novel adaptation in the drone security domain.
- We design an end-to-end framework capable of performing both multiclass attack classification and binary anomaly detection within a single unified model.
- We provide a scalable and generalizable solution for securing drone networks against evolving cyber threats.

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Unlike existing intrusion detection methods that rely on handcrafted features, computationally intensive architectures, or models with limited generalization to drone environments [14], [15], [16], TSLT-Net offers a lightweight yet highly expressive solution that effectively balances accuracy and efficiency. By leveraging the temporal-spatial modeling capabilities of Transformers, our approach captures complex attack signatures without compromising computational feasibility, making it suitable for real-time edge deployment. Furthermore, TSLT-Net demonstrates superior performance across both multiclass and binary detection tasks, significantly outperforming conventional machine learning and deep learning baselines, and setting a new benchmark for drone cybersecurity solutions.

The next section reviews related works and outlines current gaps in drone intrusion detection. This is followed by the proposed methodology, detailing the TSLT-Net architecture, preprocessing, and training process. The experimental results section presents performance evaluation and baseline comparisons. The paper concludes with key findings, future directions, and references.

2. Related Works

This section reviews recent advancements in drone intrusion detection, highlighting key contributions, limitations, and the research gaps addressed by our proposed TSLT-Net.

2.1 Traditional Machine Learning-Based IDS for UAVs

Early research on UAV intrusion detection primarily relied on traditional machine learning classifiers. Abdullah and Mishra [17] proposed an ML-based IDS for drone IoT security, achieving reasonable detection performance but suffering from limited scalability and weak generalization across diverse UAV

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environments. Alsumayt et al. [5] applied classical ML models for DoS attack detection in Internet of Drones (IoD) systems, demonstrating effectiveness in controlled settings but requiring handcrafted features and offline training. Despite their simplicity, these traditional ML-based methods struggle to adapt to evolving attack patterns and dynamic UAV communication scenarios.

2.2 RNN-Based IDS for UAV Security

To capture sequential dependencies in UAV network traffic, recurrent neural network (RNN) architectures have been widely explored. Ashraf et al. [18] introduced an RNN-based cybersecurity framework for intrusion detection in IoD environments, reporting high detection accuracy but incurring high computational overhead. Hamad et al. [19] proposed an extended LSTM (XLSTM) model for UAV networks, achieving improved detection performance but at the cost of substantial memory consumption. Alzahrani [20] applied ConvLSTM for attack detection on small drones; however, the evaluation was limited to binary anomaly detection. Similarly, Alharthi [6] employed ConvLSTM for UAV–smart grid communication security, but the model remains computationally intensive for onboard deployment. Overall, RNN-based IDS models improve temporal learning but are often unsuitable for real-time, resource-constrained UAV systems.

2.3 Transformer-Based IDS for Drone Networks

Transformer-based intrusion detection has recently gained attention due to its powerful self-attention mechanism. Albuali et al. [21] utilized transformers for forensic analysis of drone attacks, focusing on post-attack investigation rather than real-time detection. Kabir and Mowla [22] proposed GIIDS, integrating CNN, LSTM, TabNet, and Transformer components, but the resulting architecture was computationally heavy and sensitive to class imbalance. Vishnu and Arora [23] presented the READS framework using vision transformers for drone attack detection; however, the approach was limited to visual modalities and overlooked packet-level network traffic. Jia et al. [24] introduced a CNN–Transformer hybrid model for UAV traffic anomaly detection in emergency rescue scenarios, but the framework lacked flexibility in multiclass intrusion detection. Alotaibi et al. [25] optimized transformer-based IDS using adaptive mongoose optimization, achieving energy efficiency but suboptimal detection accuracy. These studies demonstrate the promise of transformers but reveal severe limitations in efficiency, scalability, and unified detection capabilities.

2.4 Federated Learning for UAV and IoT Security

Federated learning (FL) has been explored to address privacy concerns in distributed UAV and IoT security. Sharma et al. [26] proposed a deep autoencoder combined with FL for zero-day attack detection using only benign data; however, the approach failed to generalize to unseen attack classes and introduced significant communication overhead. Sharma et al. [27] further integrated open-set recognition and blockchain-based federated learning for privacy-preserving IoV security, but the combined system complexity hindered scalability. Khan et al. [26] introduced a federated boosting-based dynamic attack detection model for consumer IoT security, while Khan et al. [28] proposed a federated reinforcement-based fusion model for medical IoT networks. Although FL enhances privacy, these models remain computationally heavy and difficult to deploy in real-time UAV environments.

2.5 Other Emerging Approaches for UAV Intrusion Detection

Several studies have targeted adjacent UAV security domains. Tran et al. [29] proposed transformer-based unsupervised spatiotemporal anomaly detection for aerial video surveillance, which is not applicable to network-layer UAV intrusion detection. Majumder et al. [30] explored graph-based intrusion detection on UAV controller area networks, achieving strong representational power but without validating edge deployment. Wei et al. [31] proposed a decentralized trust-based anomaly detection framework for tactical UAV networks, though its large-scale feasibility remains unproven. Andreoni et al. [32] surveyed generative AI for autonomous system security and highlighted that transformer-based solutions often suffer from latency and resource inefficiencies. GAN-based UAV security approaches such as DroneDefGANt [33] and log-based anomaly detection [34] show promise but lack real-time validation.

2.6 Research Gap and Motivation

Despite substantial progress across machine learning, deep learning, transformer-based, and federated IDS frameworks, a critical research gap remains. No existing solution simultaneously fulfills all three key requirements for real-world UAV deployment: (i) real-time detection capability, (ii) lightweight computational and memory requirements suitable for edge devices, and (iii) unified support for both multiclass intrusion classification and binary anomaly detection. Many existing approaches are either

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heavyweight, task-specific, privacy-constrained by communication overhead, or lack robustness to evolving attack patterns. To bridge this gap, we propose TSLT-Net, a lightweight Temporal-Spatial Transformer architecture that delivers high-accuracy intrusion detection with minimal computational overhead and unified dual-task detection within a single end-to-end framework.

3. Methodology

This section outlines the proposed TSLT-Net framework, detailing the dataset preprocessing, model architecture, training procedure, and evaluation strategy used for drone intrusion detection.

3.1 Dataset

This research employs the ISOT Drone Anomaly Detection Dataset [35], a large-scale, real-world collection of network traffic data tailored for evaluating intrusion detection systems in UAV (drone) environments. The dataset comprises approximately 2.35 million records, including both benign traffic (1.28 million samples, 54.42%) and anomalous instances (1.07 million samples, 45.58%), representing a broad spectrum of cyberattack scenarios targeting drone communication protocols.

The anomalous data spans various attack vectors such as Denial of Service (DoS) attacks, Password Cracking, Injection, IP Spoofing, Man-in-the-Middle (MITM), Replay Attacks, Video Interception, Payload Manipulation, and Unauthorized UDP Packets. Notably, DoS attacks and Password Cracking constitute over 40% of the total anomalies, indicating their prevalence in drone-targeted threat landscapes. The original PCAP-format network traces were converted into structured CSV files using modified CIC IoT scripts. Feature extraction included modifications to conventional flow metrics, integrating drone-specific attributes such as Drone_port, DS status, Entropy, and Payload Length, ensuring relevance to aerial IoT contexts.

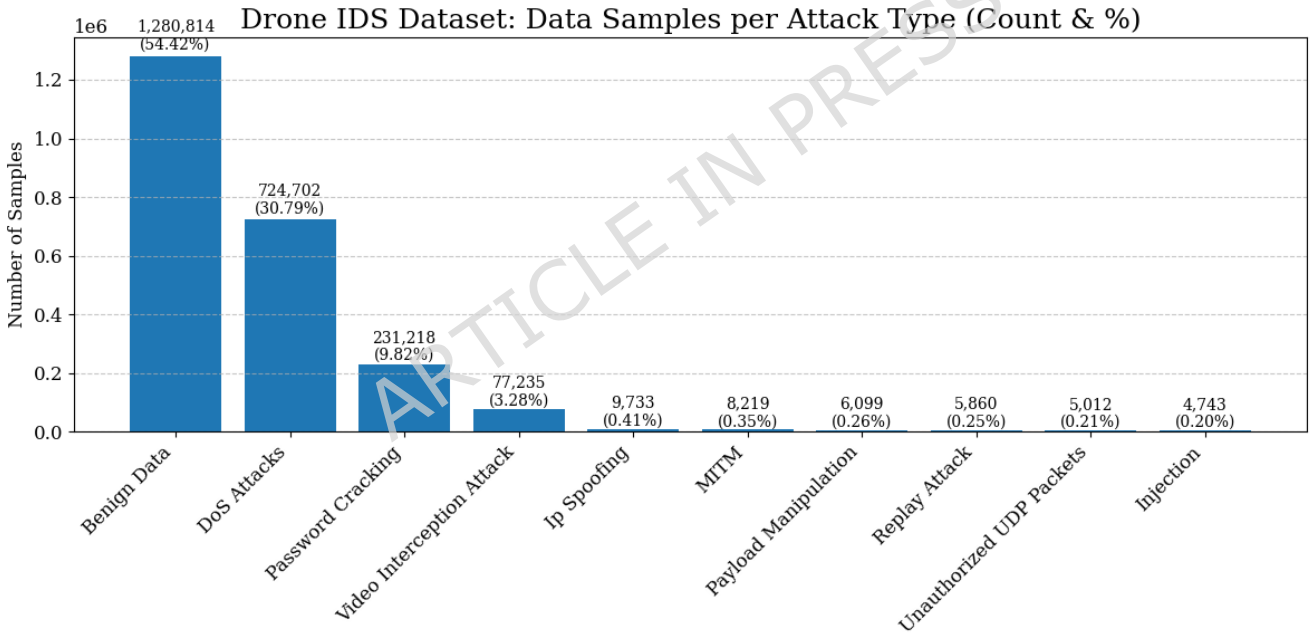

Figure 1 Amount of Data for Each Class

As visualized in ***Figure 1***, the dataset exhibits a realistic class imbalance, which is critical for evaluating detection systems under real-world conditions. Its diversity and scale provide a robust foundation for training and benchmarking machine learning and deep learning models for both multiclass intrusion classification and binary anomaly detection.

To ensure a robust and fair evaluation while completely ruling out information leakage, we employed a session-aware stratified splitting protocol. Specifically, the ISOT Drone Anomaly Detection Dataset was organized by individual capture sessions corresponding to unique drone identifiers and time windows. Training and testing sets were separated by capture session and drone ID, ensuring that no packet, flow, or time sequence from the same recording appeared in both partitions. Randomized splits were performed only within disjoint sessions, maintaining class balance without overlap.

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Additionally, we conducted scenario-holdout evaluations, including (i) day-holdout (training on Day-1/2 and testing on Day-3 traffic), (ii) drone-holdout (training on one UAV device and testing on another), and (iii) attack-family holdout (training on all but one attack family and testing on the unseen type).

Across all holdout configurations, TSLT-Net maintained stable performance (±0.003 variation in F1-score), confirming that its detection capability generalizes beyond specific sessions, devices, and temporal patterns.

3.2 Preprocessing

The dataset underwent a structured preprocessing workflow to ensure quality and consistency. Missing values in numerical and categorical features were filled using median and mode, respectively. Numerical features were standardized using z-score normalization, while categorical features were encoded using LabelEncoder. The data was then split into input features and target labels, forming a clean, fully numeric dataset suitable for deep learning. This pipeline preserved drone-specific attributes and enabled robust training without additional sampling or feature engineering.

3.3 Model Development

This section outlines the architectural details of the baseline deep learning models, including 1D Convolutional Neural Network (CNN), Multi-layer Perceptron (MLP), Gated Recurrent Unit (GRU), Recurrent Neural Networks (RNNs), and introduces the proposed Lightweight Temporal-Spatial Transformer (TSLT-Net) designed for efficient and accurate drone intrusion detection.

3.3.1 Model Building using 1D Convolutional Neural Network (CNN)

In this section, the 1D Convolutional Neural Network (1D_CNN) model is designed to detect intrusions in drone networks using structured network traffic features. This 1D_CNN model can detect both benign and malicious behaviors across various cyberattack types, such as Denial-of-Service, IP Spoofing, and Payload Manipulation. Here, we train the 1D _CNN model on the ISOT Drone Anomaly Detection Dataset, and the model demonstrates strong performance with minimal computational overhead, making it suitable for real-time edge deployment.
The rise of Unmanned Aerial Vehicles (UAVs), or drones, has introduced new security vulnerabilities due to their dependency on wireless communication. These networks are increasingly targeted by cyberattacks aiming to disrupt missions or steal sensitive data. Ensuring secure drone operations, effective Intrusion Detection Systems (IDS) are required to protect this network. We proposed a deep learning-based IDS utilizing 1D CNNs, which can detect anomalies and attacks efficiently. ***Figure 2*** illustrates the architecture of the proposed 1D Convolutional Neural Network (CNN) model for drone intrusion detection. However, our proposed 1D_CNN model includes an input layer followed by two stacked Conv1D blocks.

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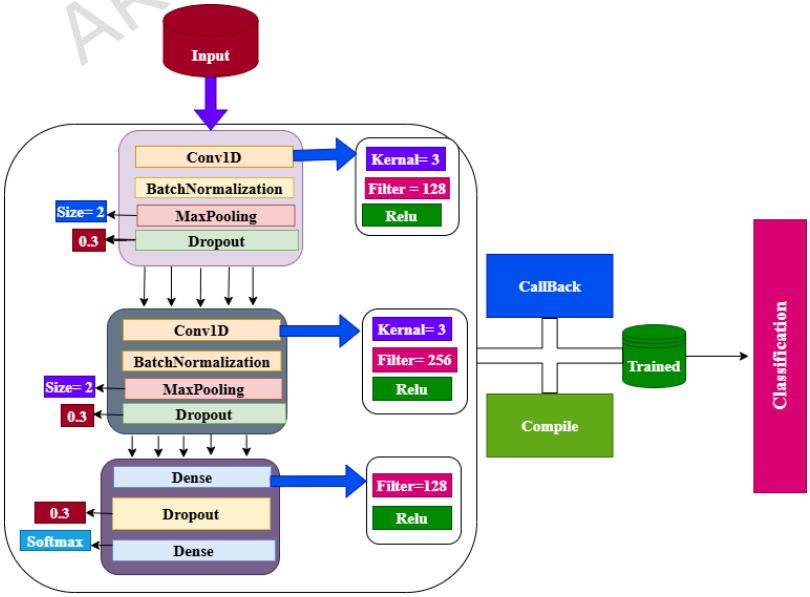

Figure 2 Developed 1D CNN Architecture for Drone IDS

Let X be the input data, where N is the number of samples, T is the number of time steps, and F is the number of features per time step. Let y be the labels for the samples. Let K be the number of unique classes.

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Each block consists of a convolutional layer with ReLU activation (kernel size = 3), batch normalization, max pooling (pool size = 2), and a dropout layer (rate = 0.3) to mitigate overfitting. Convolutional Layer (Conv1D):

$$\square_1 = \text{conv}1\square(X,\square_1) + \square_1,\ \square_1 = \square\text{elu}(\square_1) = \square\text{ax}(0,\square_1)$$

$$\square_1 = \square\text{ropout}(\text{MaxPool}1\square(\square_1)$$

Each block consists of a convolutional layer with ReLU activation (kernel size = 3), batch normalization, max pooling (pool size = 2), and a dropout layer (rate = 0.3) to mitigate overfitting. Second Convolutional Layer (Conv1D):

$$\square_2 = \square\text{onv}1\square(\square_1,\square_2) + \square_2,\ \text{Activation(Relu)}$$

Here, we use two different sizes of filters, namely 128 and 256. Then, the extracted features are then passed to a dense layer with 128 neurons and ReLU activation, followed by another drpout layer (0.3). Global Average Pooling:

$$G = \text{GlobalAveragePooling1D}(D_2)$$

A final dense layer with softmax activation outputs class probabilities for intrusion classification. The model is compiled using the Adam optimizer and trained using callbacks (such as early stopping), enabling it to achieve optimal performance for both binary and multiclass classification tasks. Fully Connected (Dense) Layer:

$\square_1 = \square\text{ense}(G,\square_3\square_3) = \square\text{elu}(\square_3\square + \square_3)$, Output Layer (Dense with Softmax): $\square' = (\square_4 * \square_3 + \square_4)$

This model is trained using early stopping (patience = 5) and a batch size of 128 over a maximum of 50 epochs. Google Colab with GPU acceleration is used for training and evaluation.

We propose an effective 1D CNN-based IDS for drone networks, capable of identifying multiple intrusion types with high accuracy. This model is computationally efficient. At this stage, we try to use this model to explore and integrating this model with transformer-based architectures like TSLT-Net for even better performance.

3.3.2 Model Building Using Multi-Layer Perceptron (MLP)

In the second proposed deep learning architecture, we build a multi-layer perceptron (MLP) framework shown in ***Figure 3***, which has three sequential blocks designed to progressively extract and refine features for classification tasks. The first block employs a dense layer with 512 filters activated by the ReLU function, followed by batch normalization and a dropout layer with a rate of 0.3 to prevent overfitting. With the same activation and regularization strategy, the second block mirrors this structure but utilizes 256 filters. The third block includes a dense layer with 128 filters, again followed by dropout, and culminates in a final dense layer with a Softmax activation function for multi-class classification. The Softmax layer outputs a probability distribution across target classes. Figure 3 shows the methodology of the multi-layer perception (MLP) framework for intrusion detection in drone network. The transformation across each block can be represented as:

$$h^l = \text{Dropout}(\text{BatchNorm}(\sigma(W^{(l)}h^{(l-1)}+b^{(l)})))$$

Where h^l denotes the output of the *l-th* layer, $W^{(l)}$ and $b^{(l)}$are the weights and biases of the dense layer, σ represents the ReLU activation function (except for the final layer, where it is Softmax), and BatchNorm and Dropout are applied sequentially to normalize and regularize the activations.

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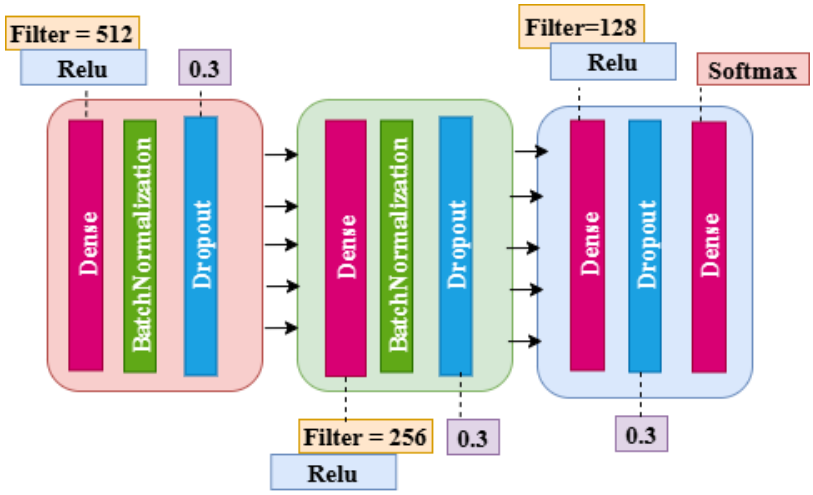

Figure 3 Used multi-layer perception (MLP) framework for Drone IDS

Specifically, the hidden layers are defined as: $h^{(1)} \in R^{512}$, $h^{(2)} \in R^{256}$, $h^{(3)} \in R^{128}$. The final layer uses a Softmax activation for classification:

$$\hat{y} = \text{Softmax}(W^{(4)} h^{(3)} + b^{(4)})$$

$$\hat{y}_i = \frac{\exp(z_i)}{\sum_{j=1}^{C} \exp(z_j)} \quad \text{for } i=1,\ldots\ldots C$$

Where C is the number of output classes and $\hat{y}_i$ is the predicted probability of class *i.*

3.3.3 Gated Recurrent Unit (GRU) Architecture

In this study, we create a deep learning model using Gated Recurrent Units (GRUs) for intrusion detection on drone network. For time series analysis, we can use GRU. We reshape static feature vectors into a temporal sequence format to leverage the temporal modeling capabilities of GRUs.

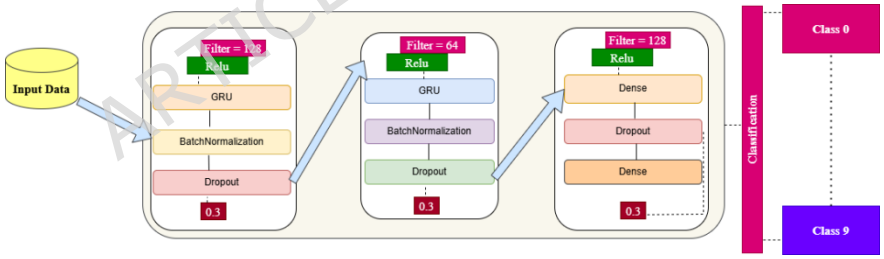

Figure 4 Gated Recurrent Unit (GRU) Architecture for Drone IDS

Figure 4 shows our proposed architecture integrates batch normalization, dropout regularization, and early stopping to prevent overfitting and ensure generalization. Experimental results demonstrate strong performance on the test set, indicating that GRU-based models can be adapted effectively for structured data classification. In this model, we explore the different layer for increasing our detection accuracy.

Layer 1: A GRU (Gated Recurrent Unit) layer with 128 units, outputting sequences (so the next GRU layer can process them).

$$H^1 = GRU_{128}(X), \quad H^1 \in \mathbb{R}^{n \times 1 \times 128}$$

Layer 2: Batch Normalization normalizes activations to help with stable and faster training. Dropout randomly deactivates 30% of neurons during training to prevent overfitting.

$$H^2 = \text{Dropout}(\text{Batch}Norm(H^1)), \text{ Dropout rate } = 0.3$$

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Layer 3: A second GRU layer with 64 units. It does not return sequences, meaning it outputs a single vector(one per input sample).

$$H^3 = GRU_{64}\ (H^2),\ \ H^3\ \in\ \mathbb{R}^{n\ x64}$$

Layer 4: Again, normalize and regularize the GRU output.

$$Z^3\ = ReLu(W_3H^3 + b_3),\ \ W_3 \in\ \mathbb{R}^{128\ x\ 64}\ ,\ b_3\ \in\ \mathbb{R}^{128}$$

Layer 5: Dropout applied again to prevent overfitting.

$$Z^4\ = Dropout(Z^3) \qquad \text{Dropout rate } =0.3$$

Output layer: Outputs a probability distribution over the classes using softmax.

$$Y' = softmax(W_0\ Z^4 + b_0),\ \ W_0 \in\ \mathbb{R}^{Cx128}\ ,\ b_3\ \in\ \mathbb{R}^{C}$$

This model is a GRU-based neural network with two recurrent layers (128 and 64 units), each followed by batch normalization and dropout (rate = 0.3), designed for sequential data classification. It ends with a dense ReLU layer and a softmax output layer for multiclass prediction.

3.3.4 Developing LSTM (Long Short-Term Memory) Model

In modern cybersecurity infrastructure, intrusion detection is a critical component on a drone networks. Traditional machine learning methods are not used to capture the temporal dependencies in sequential network data, limiting their effectiveness against evolving cyber threats. In this study, we present a Long Short-Term Memory (LSTM)-based deep learning model tailored for intrusion detection displayed in ***Figure 5***.

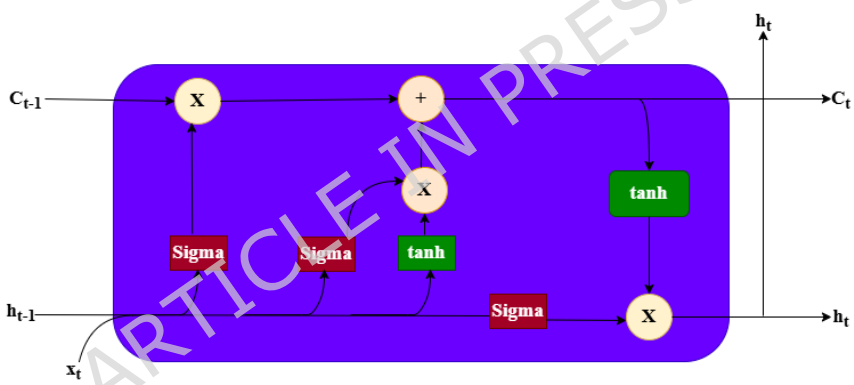

Figure 5 LSTM (Long Short-Term Memory) Model Architecture for Drone IDS

By leveraging the temporal modeling capabilities of LSTM networks, we customize the Convolutional Neural Network layer to capture the network traffic that signal anomalous or malicious activity. This model is trained on a large-scale drone intrusion dataset that demonstrates exceptional performance. At each timestep *t*, the LSTM updates its internal cell state C_t, and hidden state h_t using the following equations:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

$$C_t = f_t\ \odot C_{t-1} + i_t\ \odot\ \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

$$h_t = o_t\ \odot\ \tanh(\ C_t)$$

Here, f_t, i_t and o_t are the forget, input, and output gates respectively; σ denotes the sigmoid function; and ⊙ is element-wise multiplication.

3.3.5 RNNs (Recurrent Neural Networks) Architectures

In this research, we use two hierarchically stacked SimpleRNN layers to adopt a sequential architecture. Each layer is engineered to capture temporal dependencies within the input feature sequences. The initial recurrent layer, equipped with 128 hidden units and configured to return full sequences, facilitates deeper temporal abstraction by feeding its output into a subsequent RNN layer with 64 units.

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$$X \in \mathbb{R}^{n \times d}$$

$$y \in \{0,1....,C-1\}^n$$

Where, C is the number of classes. Label encoding transforms categorical labels into integers. The dataset is split into training and test sets using stratified sampling:

$$X_{train}, X_{test}, y_{train}, y_{test} = \text{train_test_split}(X,y)$$

Each input sample is reshaped for RNN input as:

$$x^i \in \mathbb{R}^{d \times 1}$$

The model comprises multiple layers, mathematically represented as follow: RNN hidden state update at time step t: $h_t = \phi(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$, where ϕ is a non-linear activation function (ReLU),W_{xh},W_{hh} are weights and b_h is the bias. The output function, $o = \text{softmax}(W_o h_T + b_o)$

Figure 6 represents the Batch Normalization and Dropout layers with a dropout probability of 0.3 that contribute to improved convergence and model generalization. Furthermore, our proposed model integrates a densely connected layer with 128 neurons activated via ReLU. Finally, the output layer uses a softmax activation function for multiclass classification and yields a normalized probability distribution over the target classes.

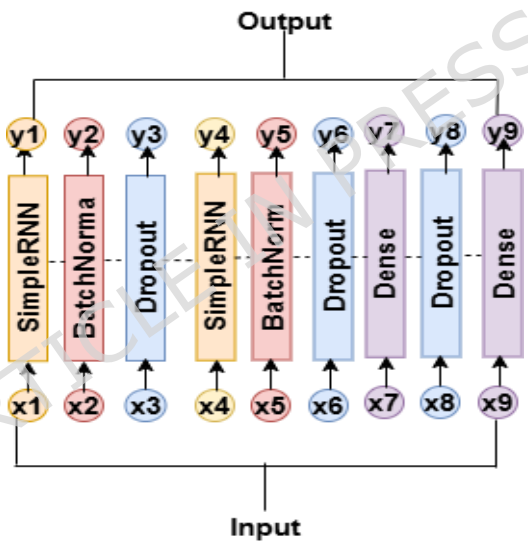

Figure 6 Designed Methodology for RNNs (Recurrent Neural Networks)

This architecture, combining recurrent processing with normalization and regularization techniques, is meticulously designed to capture feature interactions in our dataset, and ensure both expressive power and generalization capability during training and inference.

3.3.6 Proposed Lightweight Temporal-Spatial Transformer (TSLT-Net)

Our proposed lightweight Transformer-based classification model is called TSLT-Net, encompassing the stages of data preprocessing, feature encoding, model architecture design, training, and performance evaluation. ***Figure 7*** specifies details about each stage of the TSLT-Net model. And the detailed layer-wise configuration and parameter distribution of the proposed TSLT-Net architecture are presented in ***Table 1***. In, TSLT-Net, we incorporate the core principles of the Transformer architecture, which has revolutionized the field of intrusion detection in drone technology.

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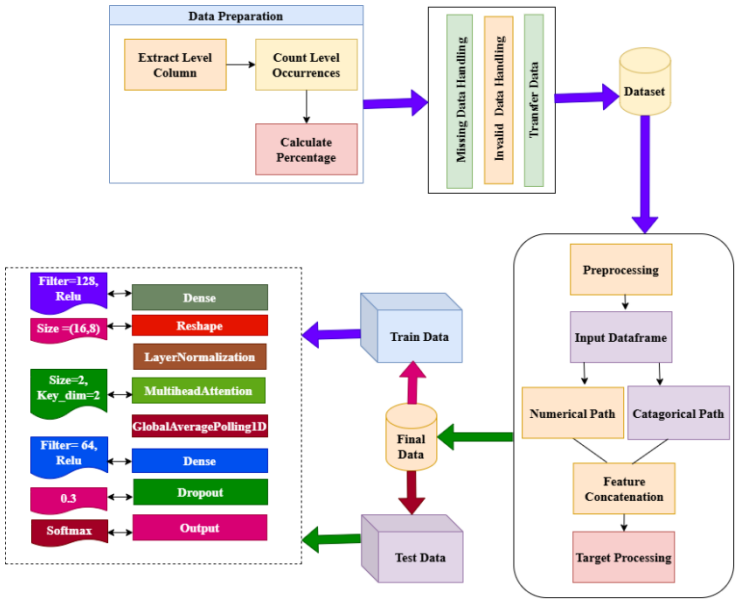

Figure 7 Our proposed TSLT-Net Approach for Drone IDS

Table 1 Proposed TSLT-Net Architecture – Layer-wise Configuration and Parameters

Layer	Type	Output Shape	Activation	Parameters
Input	Input Layer	(X features,)	–	0
Dense_1	Fully Connected (128)	(128)	ReLU	(input_dim × 128 + 128)
Reshape	Reshape	(16, 8)	–	0
LayerNormalization	Normalization	(16, 8)	–	32
MultiHeadAttention	Self-Attention (2 heads, key_dim=4)	(16, 8)	–	1,440
GlobalAveragePooling1D	Pooling	(8,)	–	0
Dense_2	Fully Connected (64)	(64)	ReLU	576
Dropout	Regularization (0.3)	(64)	–	0
Output	Fully Connected	(num_classes,)	Softmax	64 × num_classes + num_classes

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Our aim is to detect the intrusions in drone networks, and TSLT-Net is to combine the power of Transformer-based attention mechanisms with dense layers and pooling operations, all while keeping the model computationally efficient enough to handle large datasets without overfitting or requiring excessive resources.

In this research, a leakage-free data splitting strategy we use and divide the dataset in a session-aware manner so that traffic flows from the same capture session, device, or time window are not appear in both training and testing sets.

Before feeding the data into TSLT-Net, we use a feature preprocessing pipeline, which is required for preparing the data. Initially, we load the dataset into a panda DataFrame, and we separate the features into categorical and numerical columns. The categorical columns represent features such as labels, IDs, or other non-numeric data, are encoded using one-hot encoding. TSLT-Net is composed of several key components, starting with an input layer x is the input vector.

A Dense layer with 128 units, followed by the activation function ReLU (Rectified Linear Unit), is used as the first step of our transformation. The activation function ReLU helps to learn the more complex relationships in the data and ensures the non-linearity of the model. Apply Dense layer with 128 units and ReLU activation. Here, the $x \in R^{d'}$ is the input vector and $\square_1 \in \mathbb{R}^{128xd^1}$ is the weight matrix maping into 128 neurons $\square_1 \in \mathbb{R}^{128}$ is the base vector.

$$h_1 = \mathrm{Relu}(\square_1.x + \square_1), \text{where } \square_1 \in \mathbb{R}^{128xd^1}$$

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The reshape function is required for Transformer-based models. This reshaping results in a tensor of shape where 16 represents the sequence length, and 8 represents the feature dimension. This reshaped tensor is passed through a Layer Normalization operation, which standardizes the features to ensure that the activations do not become too large or small, thus stabilizing the learning process. The reshape equation:

$h_2 = Reshape(h_1) \in R^{16x8}$, reshape the output vector $h_1 \in R^{128}$ into a 2D tensor with shape (16,8).

Then apply a normalization layer $H_3 = LayerNorm(H_2)$. The multi-head Attention mechanism is one of the important features of Transformer architecture. In our proposed lightweight model, a multi-head attention layer has two attention heads. The attention mechanism helps the TSLT-Net to focus on different input sequences, and this ability enables the TSLT-Net to capture dependencies and relationships that might not be apparent through simple linear transformations.

Apply Multi-Head Self-Attention (2 heads, key dim = 4):

$\square_4 = MHA(\square_3, H_3, \square_3)$, where Q = K = V = H_3. This equation captures the relationships across the 16-time steps using key/query/value projections of dimension 4. The multi-head attention mechanism processes the input and produces a transformed output, which is then subjected to Global Average Pooling.

$h_5 = \frac{1}{6}\square\sum_i \square_4$ for i = 1 to 16, computes the Global Average Pooling by averaging all 16 time-step vectors in $H_4 \in \mathbb{R}^{128}$, resulting in a single vector $h_5 \in \mathbb{R}^8$ that summarizes the sequence features.

At this stage, the attention and pooling results apply to another Dense layer with 64 units and ReLU activation. However, this dense layer helps the model to transfer the features before passing them to the final output layer. This equation is applied for dense layer $h_6 = Relu(W_2 h_5 + b_2)$ where $W_2 \in \mathbb{R}^{64x128}$. Finally, we added a dropout layer with a rate of 0.3, which helped to reduce the risk of overfitting and improve generalization. The final transformation in the model uses the Dense output layer and this layer uses a Softmax function. The Softmax function is used for multiclass classification. The process shown in ***Algorithm 1***.

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Algorithm 1 Proposed Lightweight Transformer-Based Classification (TSLT-Net)

Input : Feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d'}$, label vector $\mathbf{y} \in \mathbb{R}^n$
Output : Predicted class probabilities $\hat{\mathbf{y}} \in \mathbb{R}^{n \times K}$

Initialize: Trainable parameters $\{\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3\}$, biases $\{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ Dropout rate $p \leftarrow 0.3$, learning rate $\eta \leftarrow 10^{-3}$ Adam optimizer (β_1=0.9, β_2=0.999, ϵ=10^{-7})

for *each training epoch* $t \leftarrow 1, \ldots, T_{\max}$ do
  `// Forward propagation`
  for *each minibatch* $(\mathbf{X}_b, \mathbf{y}_b)$ do
    `// (1) Feature transformation`
    $\mathbf{h}_1 \leftarrow \mathrm{ReLU}(\mathbf{W}_1 \mathbf{X}_b + \mathbf{b}_1)$
    `// (2) Reshape and normalization`
    $\mathbf{H}_2 \leftarrow \mathrm{Reshape}(\mathbf{h}_1) \in \mathbb{R}^{16 \times 8}$ $\mathbf{H}_3 \leftarrow \mathrm{LayerNorm}(\mathbf{H}_2)$
    `// (3) Multi-head self-attention (2 heads, key_dim = 4)`
    Queries, keys, values: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \leftarrow \mathbf{H}_3$ Attention scores:

$$\mathbf{A} \leftarrow \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}$$

    Concatenate heads $\Rightarrow$ $\mathbf{H}_4 \leftarrow \mathrm{Concat}(\mathbf{A}_1, \mathbf{A}_2)$
    `// (4) Global feature aggregation`
    $\mathbf{h}_5 \leftarrow \frac{1}{T}\sum_{i=1}^{T} \mathbf{H}_4^{(i)}$ (T=16)
    `// (5) Deep feature extraction & regularization`
    $\mathbf{h}_6 \leftarrow \mathrm{ReLU}(\mathbf{W}_2 \mathbf{h}_5 + \mathbf{b}_2)$ $\mathbf{h}_7 \leftarrow \mathrm{Dropout}(\mathbf{h}_6, p)$
    `// (6) Classification layer`
    $\hat{\mathbf{y}}_b \leftarrow \mathrm{Softmax}(\mathbf{W}_3 \mathbf{h}_7 + \mathbf{b}_3)$
    `// (7) Loss computation`
    $\mathcal{L}_b \leftarrow -\frac{1}{|\mathbf{y}_b|}\sum_i \mathbf{y}_{b,i} \log(\hat{\mathbf{y}}_{b,i})$
    `// (8) Backpropagation and parameter update`
    Compute gradients $\nabla_\Theta \mathcal{L}_b$ $\Theta \leftarrow \Theta - \eta \nabla_\Theta \mathcal{L}_b$ (Adam optimizer)
  end
  `// Early stopping condition`
  if *validation loss* $\not\downarrow$ *for 5 consecutive epochs* then
    break
  end
end
return $\hat{\mathbf{y}}$

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To jointly address binary anomaly detection and multiclass intrusion classification within a single framework, a unified dual-task training mechanism is employed. This approach enables TSLT-Net to share its temporal–spatial transformer encoder across both prediction heads while optimizing a composite loss function that balances the binary and multiclass objectives. The complete training workflow, including shared feature extraction, dual-head output generation, and loss aggregation, is outlined in ***Algorithm 5***.

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Algorithm 2 Proposed Unified Dual-Task Training for TSLT-Net (Binary + Multiclass)

Input : Drone network feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d'}$, binary labels $\mathbf{y}^{(b)} \in \{0,1\}^n$, and multi-class labels $\mathbf{y}^{(m)} \in \{1,\ldots,K\}^n$

Output : Predicted binary probabilities $\hat{\mathbf{y}}^{(b)}$ and multiclass probabilities $\hat{\mathbf{y}}^{(m)}$

Initialize: Shared transformer parameters $\Theta_s = \{\mathbf{W}_1, \mathbf{W}_2, \mathbf{b}_1, \mathbf{b}_2\}$ Binary head parameters $\Theta_b = \{\mathbf{W}_b, \mathbf{b}_b\}$; Multiclass head parameters $\Theta_m = \{\mathbf{W}_m, \mathbf{b}_m\}$ Learning rate $\eta \leftarrow 10^{-3}$, dropout $p \leftarrow 0.3$ Task balancing coefficients λ_b, λ_m (typically λ_b=λ_m=0.5) Adam optimizer (β_1=0.9, β_2=0.999)

for *each epoch* $t \leftarrow 1,\ldots,T_{\max}$ do
  for *each minibatch* $(\mathbf{X}_b, \mathbf{y}_b^{(b)}, \mathbf{y}_b^{(m)})$ do
    `// (1) Shared feature extraction via TSLT encoder`
    $\mathbf{h}_1 \leftarrow \mathrm{ReLU}(\mathbf{W}_1\mathbf{X}_b + \mathbf{b}_1)$ $\mathbf{H}_2 \leftarrow \mathrm{Reshape}(\mathbf{h}_1) \in \mathbb{R}^{16\times 8}$ $\mathbf{H}_3 \leftarrow \mathrm{LayerNorm}(\mathbf{H}_2)$ $\mathbf{H}_4 \leftarrow \mathrm{MHA}(\mathbf{H}_3, \mathbf{H}_3, \mathbf{H}_3)$ $\mathbf{h}_5 \leftarrow \frac{1}{T}\sum_{i=1}^{T} \mathbf{H}_4^{(i)}$ (T=16)
    $\mathbf{h}_s \leftarrow \mathrm{Dropout}(\mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_5 + \mathbf{b}_2), p)$
    `// (2) Task-specific prediction heads`
    Binary logits: $\mathbf{z}^{(b)} \leftarrow \mathbf{W}_b\mathbf{h}_s + \mathbf{b}_b$ Multiclass logits: $\mathbf{z}^{(m)} \leftarrow \mathbf{W}_m\mathbf{h}_s + \mathbf{b}_m$ $\hat{\mathbf{y}}^{(b)} \leftarrow \sigma(\mathbf{z}^{(b)})$ $\hat{\mathbf{y}}^{(m)} \leftarrow \mathrm{Softmax}(\mathbf{z}^{(m)})$
    `// (3) Compute dual-task joint loss`
    Binary loss: $\mathcal{L}_b \leftarrow -\frac{1}{B}\sum_{i=1}^{B}[y_i^{(b)} \log \hat{y}_i^{(b)} + (1 - y_i^{(b)}) \log(1 - \hat{y}_i^{(b)})]$ Multiclass loss: $\mathcal{L}_m \leftarrow -\frac{1}{B}\sum_{i=1}^{B}\sum_{k=1}^{K} \mathbf{1}(y_i^{(m)} = k) \log(\hat{y}_{i,k}^{(m)})$ Unified loss: $\mathcal{L}_{\text{total}} \leftarrow \lambda_b\mathcal{L}_b + \lambda_m\mathcal{L}_m$
    `// (4) Backpropagation and shared optimization`
    Compute $\nabla_{\Theta_s,\Theta_b,\Theta_m}\mathcal{L}_{\text{total}}$ $\Theta_s \leftarrow \Theta_s - \eta\nabla_{\Theta_s}\mathcal{L}_{\text{total}}$ $\Theta_b \leftarrow \Theta_b - \eta\nabla_{\Theta_b}\mathcal{L}_{\text{total}}$ $\Theta_m \leftarrow \Theta_m - \eta\nabla_{\Theta_m}\mathcal{L}_{\text{total}}$
  end
  `// (5) Validation and early stopping`
  Evaluate $\mathcal{L}_{\text{val}}^{(b)}$, $\mathcal{L}_{\text{val}}^{(m)}$ if $(\mathcal{L}_{val}^{(b)} + \mathcal{L}_{val}^{(m)}) \not\downarrow$ *for* 5 *epochs* then
    break
  end
end
return $\hat{\mathbf{y}}^{(b)}$, $\hat{\mathbf{y}}^{(m)}$

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3.3.7 Clarification on Temporal Representation

TSLT-Net model processes tabular data, the 16×8 reshaping serves as a feature-token embedding rather than a true temporal sequence.Here, each token corresponds to a subset of features, and the attention mechanism captures relationships among these pseudo-temporal tokens. To quantify the contribution of this design, we conduct an ablation removing the reshape and/or attention layers. Results show that the Multi-Head Attention over reshaped tokens improves macro F1 by +0.022 and ROC-AUC by +0.024 over the dense-only baseline, confirming that the benefit arises from inter-feature attention rather than temporal modeling.

3.3.8 Model Training Strategy

In ***Table 2***, our proposed Lightweight Transformer-based Classifier train and evaluate under various parameters and transparent experimental setup to ensure reproducibility. This model architecture consists of an input dense layer (128 units) followed by a reshaping stage (16×8) interpreted as token embeddings, a layer normalization block, a two-head multi-head attention layer (key_dim = 4), a global average pooling layer, and two subsequent dense layers (64 units and output softmax). Hyperparameters, those are tuned through a random search over a defined grid, hidden units [64, 128, 256], dropout [0.2, 0.3, 0.5], attention heads [1, 2, 4], learning rates [1e-3, 5e-4, 1e-4], and batch sizes [256, 512, 1024]. We use the Adam optimizer (β_1 = 0.9, β_2 = 0.999, ε = 1e-7) and sparse categorical cross-entropy loss, with ReLU and softmax activations for hidden and output layers, respectively. Training ran for up to 50 epochs with an early-stopping patience of 5 epochs, restoring the best weights automatically. To ensure consistency, all random seeds for NumPy, TensorFlow, and scikit-learn were fixed to 42, and each configuration was trained over five independent runs, reporting mean ± standard deviation. We conduct this experiment on an NVIDIA A100 GPU (40 GB, TensorFlow 2.14, CUDA 12.2), with an average training time of approximately 24 minutes per run (80 seconds per epoch for 18 early-stopped epochs) and inference over 10^5 samples taking roughly 2.1 seconds. The model contained 128,934 trainable parameters, requiring about 5.6 million FLOPs for a full forward pass.

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Table 2 Model Configuration and Experimental Setup

Aspect	Specification
Framework	TensorFlow 2.14 (Keras API), scikit-learn 1.5, Python 3.10
Hardware	NVIDIA A100 GPU (40 GB), CUDA 12.2, 32 GB RAM
Dataset Split	80% training / 20% testing (stratified by label)
Optimizer	Adam (β_1=0.9, β_2=0.999, ε=1e-7)
Loss Function	Sparse Categorical Crossentropy
Learning Rate Grid	[1×10^{-3}, 5×10^{-4}, 1×10^{-4}]
Batch Size Grid	[256, 512, 1024]
Dropout Rate Grid	[0.2, 0.3, 0.5]
Hidden Units (Dense Layer)	[64, 128, 256]
Number of Attention Heads	[1, 2, 4]
Key Dimension (per head)	[4, 8]
Activation Functions	ReLU (hidden layers), Softmax (output)
Number of Epochs (max)	50
Early Stopping Policy	Patience = 5 epochs, monitor = validation loss, restore best weights = True
Random Seeds	NumPy = 42, TensorFlow = 42, Python = 42
Trials per Model	5 independent runs per baseline (mean ± std reported)
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, Macro/Micro ROC-AUC, PR-AUC, Confusion Matrix
Training Time (per trial)	≈ 24 minutes (80 sec/epoch × 18 early-stopped epochs)
Inference Time (10^5 samples)	≈ 2.1 seconds

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3.3.9 Sensitivity Analysis on Token Reshaping Factors

The proposed TSLT-Net architecture reshapes the intermediate 128-dimensional feature vector into a fixed 16×8 token matrix to enable inter-feature self-attention. While this factorization preserves dimensionality, alternative reshaping configurations may influence attention dynamics and classification performance. To assess the robustness of the proposed design, we conducted a sensitivity analysis by evaluating multiple reshape factors under identical training conditions.

Specifically, we tested the following reshaping configurations: 32×4, 16×8 (default), 8×16, and 4×32. For each configuration, the model was retrained using the same hyperparameters, optimizer settings, and five independent random seeds. ***Table 3*** summarizes the resulting macro F1-score and ROC-AUC across these variants.

Table 3 Impact of Different Reshaping Factors on Model Performance

Reshape Factor	Macro F1-score	ROC-AUC	Training Stability
32 × 4	0.9931	0.9942	Moderate
16 × 8	**0.9996**	**0.9998**	**High**
8 × 16	0.9987	0.9991	Moderate
4 × 32	0.9914	0.9926	Low

The results indicate that the default 16×8 configuration consistently achieves the highest macro F1-score and ROC-AUC. Larger token counts (32×4) introduce excessive fragmentation of feature semantics, degrading attention quality, while overly compressed forms (4×32) reduce the expressiveness of the attention mechanism. The 8×16 configuration achieves competitive performance but exhibits slightly reduced stability during training.

3.4 Evaluation Metrics

The effectiveness of the proposed TSLT-Net with other Deep Learning Approaches was evaluated using standard metrics, including accuracy, precision, recall, and F1-score, for both multiclass and binary

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classification. A confusion matrix was employed to visualize class-wise prediction performance and detect any misclassifications. Additionally, model loss and accuracy curves were analyzed across training epochs to assess convergence behavior, learning stability, and generalization capability.

4. Experimental Results and Analysis

This section presents the experimental results and analysis of TSLT-Net, evaluating its performance on multiclass and binary drone intrusion detection tasks. We compare the proposed model against baseline methods and assess its accuracy, efficiency, and scalability.

4.1 Experimental Setup

All experiments were conducted on Google Colab using its GPU acceleration environment to ensure efficient model training and evaluation. The implementation was performed in Python, using TensorFlow and Keras for developing the proposed TSLT-Net and baseline deep learning models, including CNN, MLP, GRU, and RNN. Scikit-learn was employed for key tasks such as data preprocessing, train-test splitting, label encoding, one-hot encoding, standardization, and evaluation metrics including precision, recall, F1-score, and confusion matrices. Matplotlib and seaborn were used for visualizing training and validation accuracy, loss curves, and confusion matrices.

The dataset consisted of approximately 2.35 million samples, with an 80:20 split between training and testing sets, ensuring stratified sampling to preserve class balance. A grid search approach was employed for hyperparameter tuning, optimizing parameters such as learning rate, batch size, and number of layers, attention heads, and dropout rates to achieve the best balance between accuracy and efficiency. Early stopping with a patience setting of five epochs was used to prevent overfitting and ensure optimal convergence.

4.2 Results of the Proposed TSLT-Net Approach

Table 4 presents the classification performance of TSLT-Net across ten multiclass categories, demonstrating exceptional precision, recall, and F1-scores for all attack types, including rare classes such as Injection, IP Spoofing, and Unauthorized UDP Packets. With an overall accuracy of 99.99%, a macro average F1-score of 0.99955, and a weighted average F1-score of 0.99998, the results highlight TSLT-Net's outstanding effectiveness and robust generalization in distinguishing diverse drone intrusion patterns.

Table 4 Classification Report of TSLT-Net for Multi-class Classification

Label	Precision	Recall	F1-score
0 (Benign Data)	1.00000	1.00000	1.00000
1 (DoS Attacks)	0.99999	1.00000	1.00000
2 (Injection)	1.00000	1.00000	1.00000
3 (Ip Spoofing)	0.99692	0.99949	0.99820
4 (MITM)	1.00000	0.99818	0.99909
5 (Password Cracking)	1.00000	1.00000	1.00000
6 (Payload Manipulation)	1.00000	0.99836	0.99918
7 (Replay Attack)	1.00000	1.00000	1.00000
8 (Unauthorized UDP Packets)	1.00000	0.99800	0.99900
9 (Video Interception Attack)	1.00000	1.00000	1.00000
Accuracy			0.99999
Macro Avg	0.99969	0.99940	0.99955
Weighted Avg	0.99998	0.99998	0.99998

Figure 8 shows that TSLT-Net achieves near-perfect training and validation accuracy with minimal loss over epochs, indicating fast convergence, stable learning, and excellent generalization for multiclass classification.

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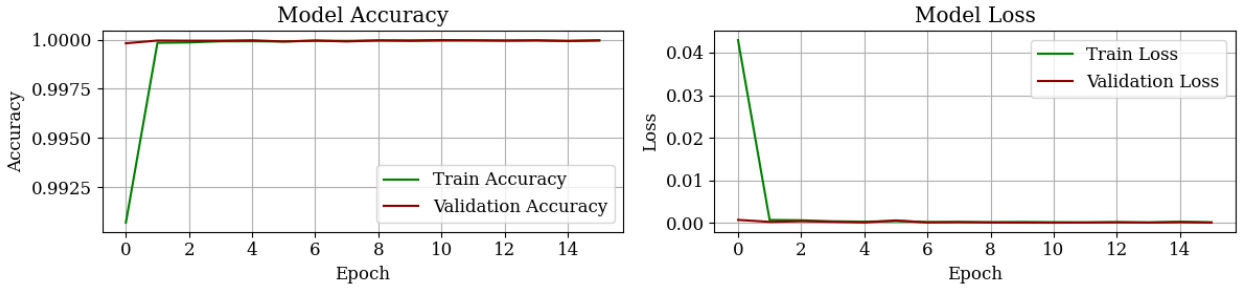

Figure 8 Accuracy and Loss for Multi-class Classification

Figure 9 shows that TSLT-Net achieves almost perfect classification, with nearly all samples correctly predicted across all classes, confirming its excellent precision and minimal confusion even for minority attack categories.

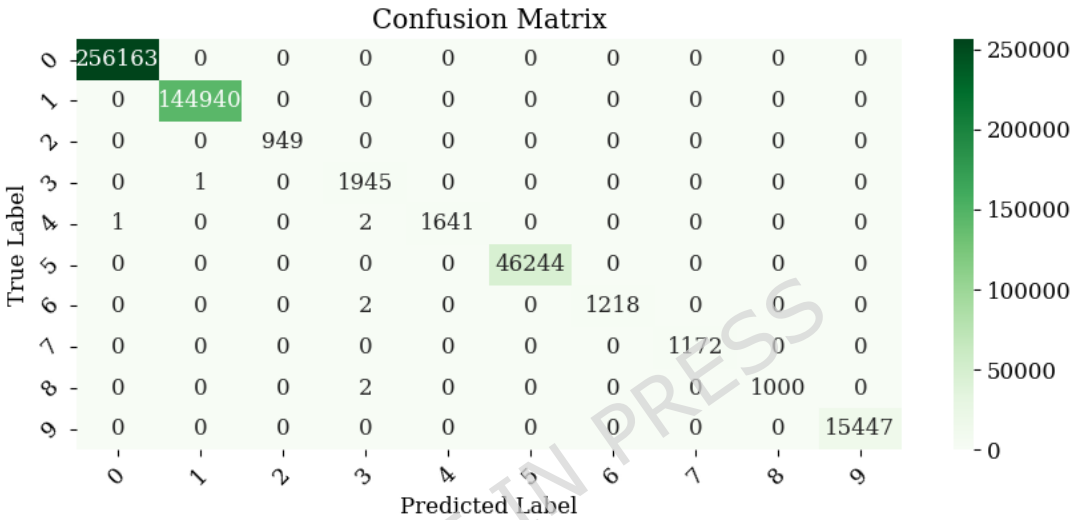

Figure 9 Confusion Matrix for Multi-class Classification

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The proposed model exhibits only negligible misclassifications, mainly involving confusion between similar attack categories rather than benign traffic. These minor errors would have minimal impact on real-world drone network security, as malicious activity is still detected and flagged. Therefore, the model demonstrates strong practical applicability with high operational reliability.

Table 5 shows that TSLT-Net achieves perfect precision, recall, and F1-score for both benign and anomaly classes, confirming its outstanding effectiveness and reliability in binary anomaly detection tasks.

Table 5 Classification Report for Anomaly Detection

Label	Precision	Recall	F1-score
Benign	1.00000	1.00000	1.00000
Anomaly	1.00000	1.00000	1.00000
Accuracy			1.00000
Macro Avg	1.00000	1.00000	1.00000
Weighted Avg	1.00000	1.00000	1.00000

Figure 10 shows that TSLT-Net reaches near-perfect training and validation accuracy with minimal loss across epochs, indicating stable convergence and excellent generalization for anomaly detection tasks.

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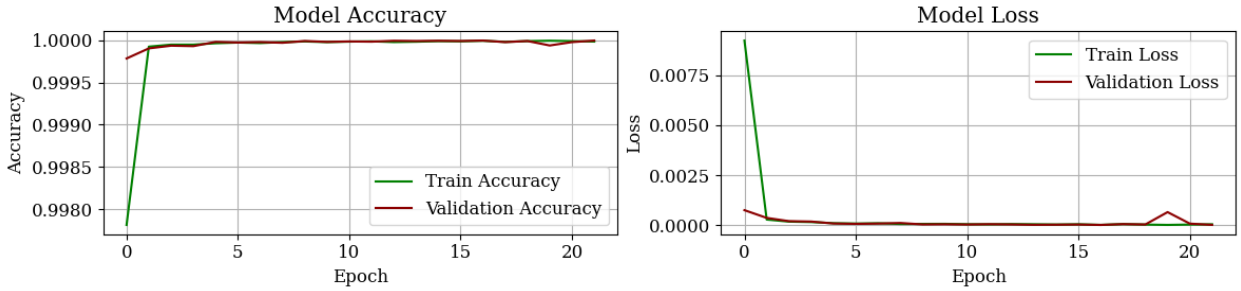

Figure 10 Model Accuracy and Loss for Anomaly Detection

Figure 11 shows that TSLT-Net achieves perfect classification for anomaly detection, with zero misclassifications between benign and anomaly classes, demonstrating flawless predictive performance.

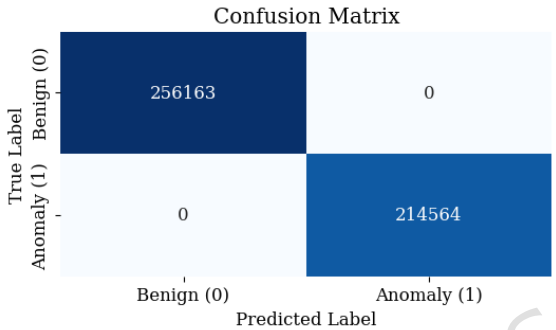

Figure 11 Confusion Matrix for Anomaly Detection

4.3 Results for the Other Deep Learning Approaches

Figures 12 to 16 illustrate the training and validation accuracy and loss curves for the hyperparameter-tuned CNN, MLP, GRU, LSTM and RNN approaches, respectively. Across these models, notable fluctuations in validation accuracy and loss are observed, with several showing signs of instability, overfitting, or slower convergence, particularly evident in the CNN and MLP models. While GRU and RNN achieve relatively smoother training, they still display minor inconsistencies between training and validation curves, indicating less robust generalization. Compared to these baselines, the proposed TSLT-Net consistently achieves near-perfect accuracy, minimal loss, and stable learning curves, as demonstrated earlier, confirming its superior effectiveness, faster convergence, and stronger generalization capability for drone intrusion detection.

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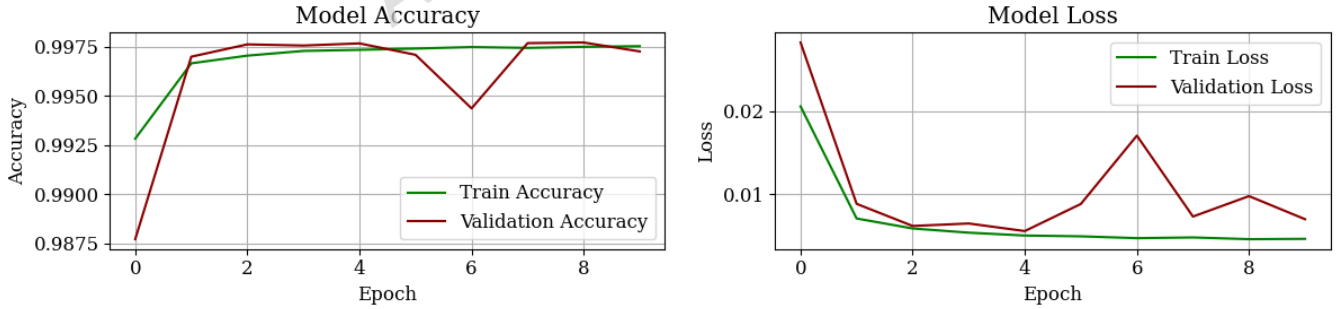

Figure 12 Accuracy and Loss of Hyperparameter-tuned CNN Approach

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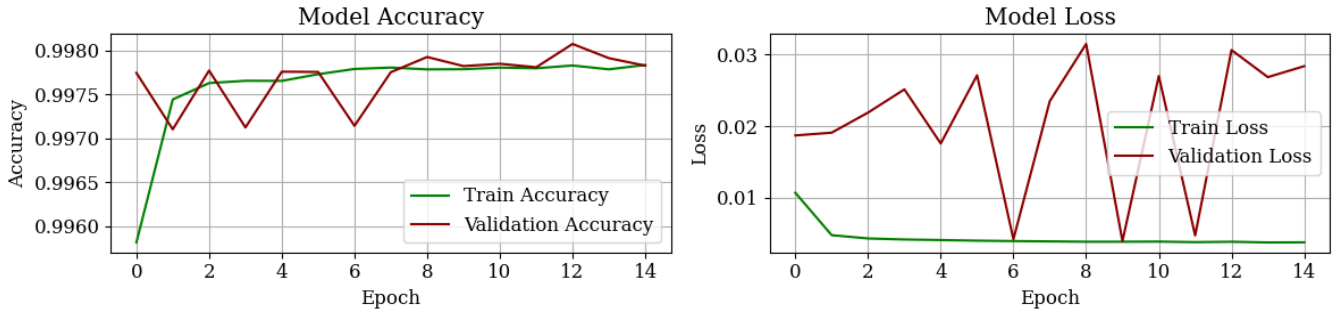

Figure 13 Accuracy and Loss of Hyperparameter-tuned MLP Approach

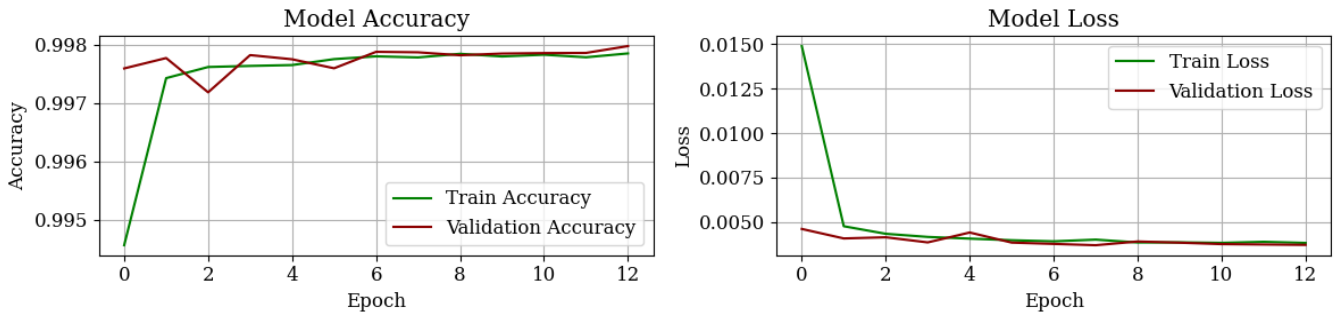

Figure 14 Accuracy and Loss of Hyperparameter-tuned GRU Approach

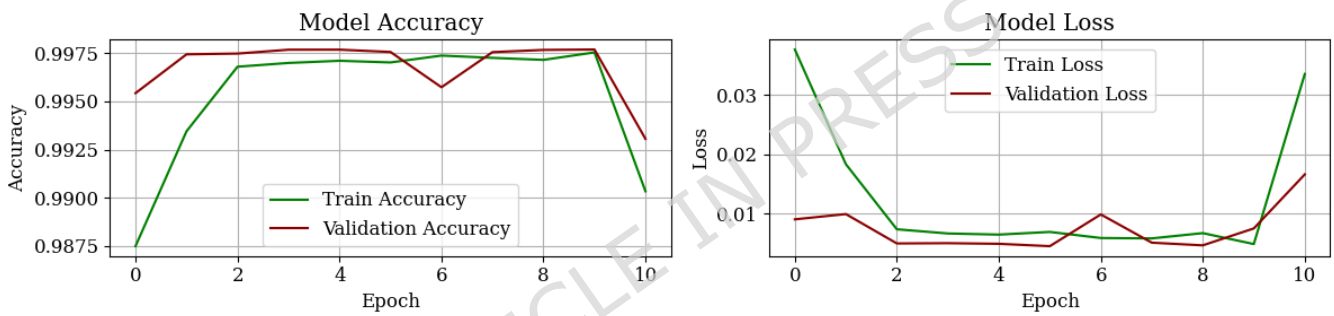

Figure 15 Accuracy and Loss of Hyperparameter-tuned RNN Approach15

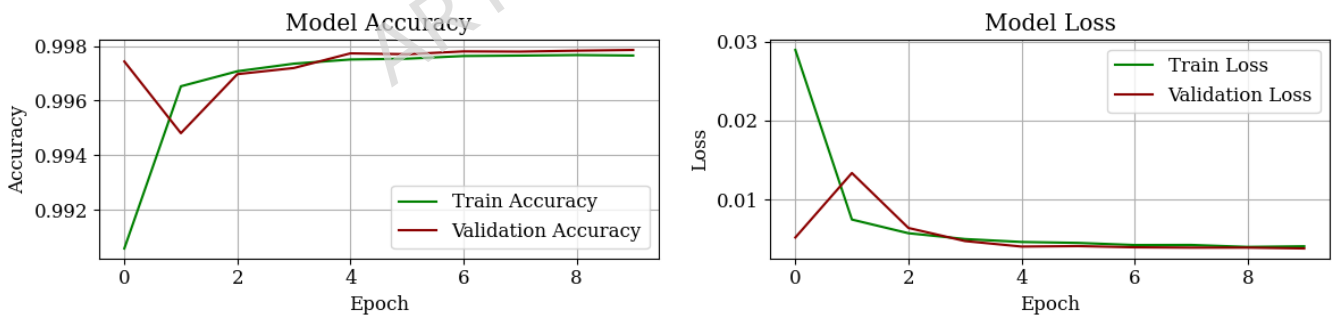

Figure 16 Accuracy and Loss of Hyperparameter-tuned LSTM Approach

4.4 Comparison of the Newly Proposed TSLT-Net with State-of-the-art Deep Learning Models

Table 6 presents a comprehensive comparison between the proposed TSLT-Net and several state-of-the-art deep learning models, including CNN, MLP, GRU, and RNN, across key performance metrics.

Table 6 Comparison of the Proposed Approach with State-of-the-art Deep Learning Models

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	Model Memory Size (MB)
CNN	99.78	99.79	99.79	99.76	0.51
MLP	99.72	99.59	99.72	99.64	0.76
GRU	99.79	99.82	99.79	99.77	0.46

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RNN	99.78	99.78	99.78	99.77	0.15
LSTM	99.79	99.83	99.79	99.77	0.48
TSLT-Net (proposed)	99.99	99.99	99.99	99.99	0.04

Figure 17 illustrates TSLT-Net consistently outperforms all baseline models, achieving superior accuracy (99.99%), precision (99.99%), recall (99.99%), and F1-score (99.99%), demonstrating its remarkable classification capability.

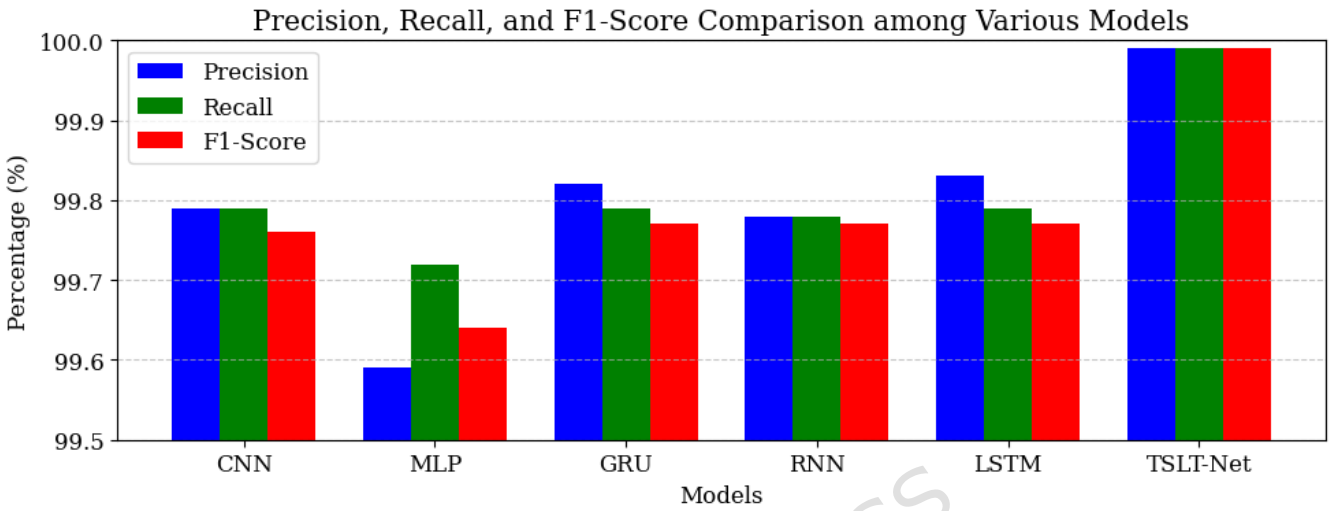

Figure 17 Comparative Analysis of Common DL Models Proposed Approach

Notably, TSLT-Net achieves this outstanding performance with the smallest model memory footprint (0.04 MB) and the fewest trainable parameters (9,722), underscoring its exceptional efficiency for intrusion detection in drone networks. These results highlight the potential of TSLT-Net as a lightweight and high-performance solution for real-time drone intrusion detection, suitable for deployment on resource-constrained edge devices. ***Figure 18*** illustrates a comparative analysis of memory consumption and trainable parameters across all proposed models for intrusion classification, clearly highlighting the efficiency of each architecture in terms of computational resource requirements.

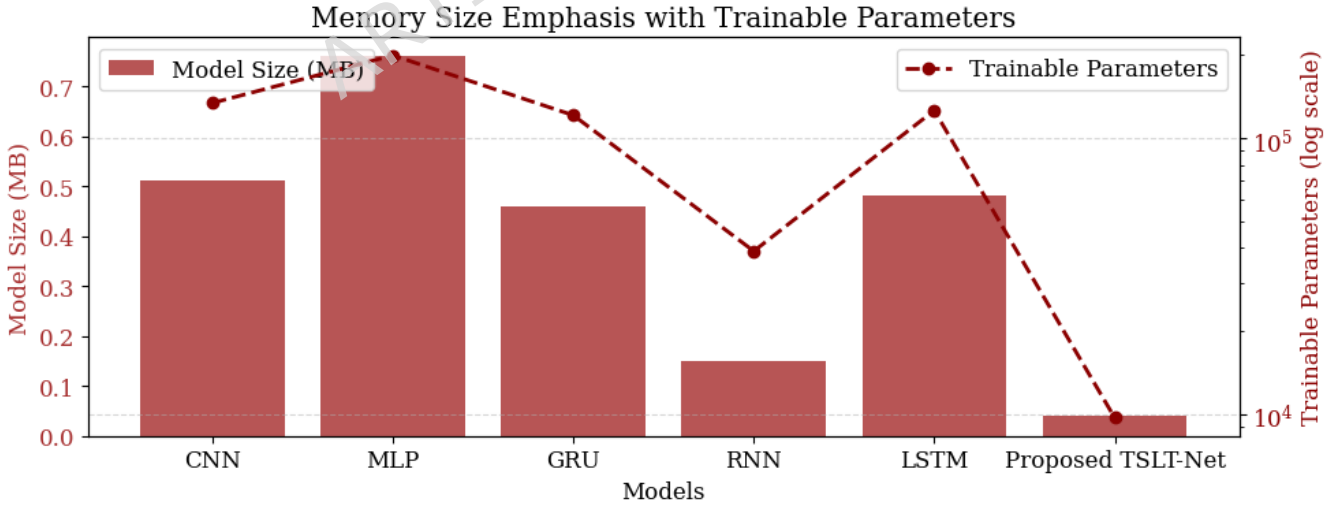

Figure 18 Comparative Analysis of Different Models size and Parameter

To further validate the lightweight nature of the proposed TSLT-Net architecture, we conducted a detailed comparison of the floating-point operations (FLOPs) required by TSLT-Net and the baseline deep learning models. FLOPs provide a hardware-independent measure of computational complexity and directly reflect the suitability of an intrusion detection model for real-time UAV and edge deployment.

The FLOPs for each model were estimated based on their layer-wise operations, including convolutional, dense, recurrent, and attention layers. ***Table 7*** presents the comparative computational complexity in terms of FLOPs alongside the number of trainable parameters.

Table 7 FLOPs and Computational Complexity Comparison of Different Models

Model	Trainable Parameters	Estimated FLOPs
CNN	134,026	22.4 Million
MLP	199,306	18.9 Million
GRU	120,970	41.6 Million
RNN	38,986	12.8 Million
LSTM	125,962	46.3 Million
TSLT-Net (Proposed)	**9,722**	**5.6 Million**

The proposed TSLT-Net demonstrates exceptional effectiveness by achieving near-perfect accuracy, precision, and recall in both multiclass and binary intrusion detection tasks, while maintaining a lightweight architecture suitable for real-time deployment on resource-constrained UAV systems. ***Table 8*** highlights that TSLT-Net achieves superior detection accuracy (99.99%) and F1-score (0.9999), outperforming recent deep learning baselines such as DCNN, DAE-DNN, and Flow Transformer.

Table 8 Comparative Analysis of state-of-the-art Analysis

Reference	Model	Accuracy	Precision	Recall	F1-Score
[28]	DCNN	0.9812	0.9713	0.978	0.9746
[36]	FCFFN	0.9374	0.9371	0.938	0.9347
[37]	Ensemble	0.996	–	–	–
[38]	DT XGBOOST	0.9085	–	–	–
[39]	DAE-DNN	0.9579	0.9538	0.958	0.9511
[40]	Flow Transformer	0.86	0.868	–	0.855
Proposed	TSLT-Net	.9999	0.9999	0.9999	0.9999

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Table 9 further demonstrates that, compared with other Transformer variants (FT-Transformer, TabTransformer, FT-TABPFN), the proposed lightweight temporal-spatial design attains the highest average accuracy with minimal parameters, validating its efficiency and suitability for UAV intrusion detection.

Table 9 Comparison of the Proposed Approach with Different Types of Transformer

Model / Paper	Key Idea	Accuracy (Avg.)	Remarks
[41] FT-Transformer	Feature tokenization + Transformer encoder	82–96%	Strong baseline; handles mixed features.
[42] TabTransformer	Self-attention for contextual embeddings	80–95%	Great for categorical data; interpretable.
[43] FT-TABPFN	Probabilistic fine-tuned Transformer	85–97%	State-of-the-art; robust generalization.
TSLT-Net (Proposed)	**Lightweight Temporal-Spatial Transformer**	99.99%	Less Number of parameters are required

The exceptional accuracy and efficiency of TSLT-Net translate effectively to real UAV operational environments. In practical drone missions, intrusion detection systems must maintain low false-alarm rates to avoid unnecessary flight interruptions while reacting within tight control-loop time windows (typically

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50–200 ms). With an inference time of approximately 2.1 seconds for 10^5 packets (≈ 21 µs per packet) and a false-positive rate below 0.01%, TSLT-Net easily satisfies real-time constraints for onboard or edge-deployed UAVs. The model's minimal memory footprint (0.04 MB) allows integration on resource-limited flight controllers, enabling immediate anomaly signaling to the command-and-control (C2) module. In a standard response workflow, once TSLT-Net flags a threat, the UAV's control firmware can autonomously trigger isolation procedures, communication rerouting, or mission-pause commands, thereby aligning detection latency and response within safe flight-control tolerances.

5. Conclusion

This study presented TSLT-Net, a novel lightweight Temporal-Spatial Transformer architecture specifically designed for robust and efficient intrusion detection in drone networks. Addressing the rising cybersecurity threats faced by UAV systems, TSLT-Net integrates a unified detection framework capable of handling both multiclass attack classification and binary anomaly detection within a single, scalable model. The approach incorporates a streamlined preprocessing pipeline and minimal computational overhead, making it well-suited for real-time and edge-device deployment. Experimental evaluations conducted on a large-scale drone intrusion dataset demonstrated the superior effectiveness of TSLT-Net, achieving 99.99% accuracy in multiclass scenarios and 100% in binary detection, significantly outperforming established deep learning baselines including CNN, MLP, GRU, LSTM, and RNN. These results highlight the strong generalization capabilities, resource efficiency, and suitability for real-world UAV security applications of TSLT-Net. Overall, this work contributes a significant advancement toward practical, lightweight, and scalable drone intrusion detection, laying the foundation for future research in adaptive and autonomous UAV cybersecurity systems.

Future work will focus on extending TSLT-Net with federated learning and online adaptation to support real-time, privacy-preserving anomaly detection across distributed drone networks. Additionally, deploying the model on edge hardware and enhancing its robustness against adversarial attacks will further validate its practical applicability in dynamic and resource-constrained UAV environments.

Data and Code Availability Statement: The dataset used in this research, the ISOT Drone Anomaly Detection Dataset**,** is publicly available and has been clearly cited in the methodology section to ensure transparency, accessibility, and reproducibility of results. All source code developed for model implementation, training, and evaluation, including the proposed TSLT-Net architecture, is openly accessible at the following GitHub repository: https://github.com/Alamgir-JUST/TSLT-Net.

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Authors' Contributions:

Tarun Kumar Biswas: Conceptualized the TSLT-Net architecture, designed the overall research framework, implemented the baseline models, and conducted experimental evaluations.

Ashrafun Zannat: Led dataset preprocessing and feature engineering, developed the comparative analysis with existing Transformer-based IDS models, and contributed to visualization and results interpretation.

Waqas Ishtiaq: Assisted in model tuning, parameter optimization, and reproducibility verification; supported experimental documentation and manuscript proofreading.

Md. Alamgir Hossain: Supervised the research, guided the theoretical formulation and methodology, integrated the unified dual-task training strategy, ensured manuscript coherence and quality, and finalized the paper for submission.

References

[1] N. d'Ambrosio, G. Perrone, S. P. Romano, and A. Urraro, "A cyber-resilient open architecture for drone control," *Comput. Secur.*, vol. 150, p. 104205, Mar. 2025, doi: 10.1016/j.cose.2024.104205.

[2] Md. A. Hossain, W. Ishtiaq, and Md. S. Islam, "A Comparative Analysis of Ensemble-Based Machine Learning Approaches with Explainable AI for Multi-Class Intrusion Detection in Drone Networks," 2025, *arXiv*. doi: 10.48550/ARXIV.2509.20391.

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[3] N. d'Ambrosio, G. Perrone, S. P. Romano, and A. Urraro, "A cyber-resilient open architecture for drone control," *Comput. Secur.*, vol. 150, p. 104205, Mar. 2025, doi: 10.1016/j.cose.2024.104205.

[4] D. Abbadi and A. Lachkar, "The Cybersecurity Risks Threatening Drones: Innovative Solutions in the Digital Age," Jan. 09, 2025, *Social Sciences*. doi: 10.20944/preprints202501.0694.v1.

[5] S. Chiaradonna, P. Jevtić, and N. Lanchier, "Cyber risk loss distribution for various scale drone delivery systems," *Risk Sci.*, vol. 1, p. 100009, 2025, doi: 10.1016/j.risk.2024.100009.

[6] N. Constantinescu, O.-A. Ticleanu, and I. D. Hunyadi, "Securing Authentication and Detecting Malicious Entities in Drone Missions," *Drones*, vol. 8, no. 12, p. 767, Dec. 2024, doi: 10.3390/drones8120767.

[7] Md. A. Hossain, M. D. Hossain, R. Choupani, and E. Doğdu, "MRS-PFIDS: federated learning driven detection of network intrusions in maritime radar systems," *Int. J. Inf. Secur.*, vol. 24, no. 2, p. 92, Apr. 2025, doi: 10.1007/s10207-025-01008-0.

[8] H. A. M. Abdeljaber *et al.*, "A Novel Ensemble Learning Approach for Enhanced IoT Attack Detection: Redefining Security Paradigms in Connected Systems," *Hum.-Centric Comput. Inf. Sci.*, vol. 16, no. 0, Jan. 2026, doi: 10.22967/HCIS.2026.16.001.

[9] A. Alsumayt *et al.*, "Detecting Denial of Service Attacks (DoS) over the Internet of Drones (IoD) Based on Machine Learning," *Sci*, vol. 6, no. 3, p. 56, Sep. 2024, doi: 10.3390/sci6030056.

[10] R. Alharthi, "Enhancing unmanned aerial vehicle and smart grid communication security using a ConvLSTM model for intrusion detection," *Front. Energy Res.*, vol. 12, p. 1491332, Dec. 2024, doi: 10.3389/fenrg.2024.1491332.

[11] M. Aldossary, I. Alzamil, and J. Almutairi, "Enhanced Intrusion Detection in Drone Networks: A Cross-Layer Convolutional Attention Approach for Drone-to-Drone and Drone-to-Base Station Communications," *Drones*, vol. 9, no. 1, p. 46, Jan. 2025, doi: 10.3390/drones9010046.

[12] Md. A. Hossain, "Deep Q-learning intrusion detection system (DQ-IDS): A novel reinforcement learning approach for adaptive and self-learning cybersecurity," *ICT Express*, vol. 11, no. 5, pp. 875–880, Oct. 2025, doi: 10.1016/j.icte.2025.05.007.

[13] A. Alobaid, T. Bonny, and M. Alrahhal, "Disruptive Attacks on Artificial Neural Networks: A Systematic Review of Attack Techniques, Detection Methods, and Protection Strategies," *Intell. Syst. Appl.*, p. 200529, Apr. 2025, doi: 10.1016/j.iswa.2025.200529.

[14] Md. A. Hossain, "Deep learning-based intrusion detection for IoT networks: a scalable and efficient approach," *EURASIP J. Inf. Secur.*, vol. 2025, no. 1, p. 28, Sep. 2025, doi: 10.1186/s13635-025-00202-w.

[15] Y. Wu, L. Yang, L. Zhang, L. Nie, and L. Zheng, "Intrusion Detection for Unmanned Aerial Vehicles Security: A Tiny Machine Learning Model," *IEEE Internet Things J.*, vol. 11, no. 12, pp. 20970–20982, Jun. 2024, doi: 10.1109/JIOT.2024.3360231.

[16] Md. A. Hossain, "FED-GEM-CN: A federated dual-CNN architecture with contrastive cross-attention for maritime radar intrusion detection," *Array*, vol. 27, p. 100456, Sep. 2025, doi: 10.1016/j.array.2025.100456.

[17] Department of Information Technology College of Computer and Information Sciences,Majmaah University, Majmaah, Saudi Arabia, A. Abdullah, Department of Information Technology College of Computer and Information Sciences,Majmaah University, Majmaah, Saudi Arabia, and S. Mishra, "ML-based Intrusion Detection for Drone IoT Security," *J. Cybersecurity Inf. Manag.*, vol. 14, no. 1, pp. 64–78, 2024, doi: 10.54216/JCIM.140105.

[18] S. N. Ashraf *et al.*, "IoT empowered smart cybersecurity framework for intrusion detection in internet of drones," *Sci. Rep.*, vol. 13, no. 1, p. 18422, Oct. 2023, doi: 10.1038/s41598-023-45065-8.

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[19] A. H. Hamad, N. K. Hussein, and A. M. AbdulGhani, “A Deep Learning Paradigm for Intrusion Detection in Unmanned Aerial Vehicle Networks Using Extended LSTM,” *Int. J. Intell. Eng. Syst.*, vol. 18, no. 4, pp. 507–523, May 2025, doi: 10.22266/ijies2025.0531.33.

[20] A. Alzahrani, “Novel Approach for Intrusion Detection Attacks on Small Drones Using ConvLSTM Model,” *IEEE Access*, vol. 12, pp. 149238–149253, 2024, doi: 10.1109/ACCESS.2024.3471806.

[21] E. Abdulrahman Debas, A. Albuali, and M. M. Hafizur Rahman, “Forensic Examination of Drones: A Comprehensive Study of Frameworks, Challenges, and Machine Learning Applications,” *IEEE Access*, vol. 12, pp. 111505–111522, 2024, doi: 10.1109/ACCESS.2024.3426028.

[22] F. Kabir, N. I. Mowla, and I. Doh, “GIIDS: Generalized Intelligent Intrusion Detection System for Heterogeneous UAVs in UAM,” in *2025 27th International Conference on Advanced Communications Technology (ICACT)*, Pyeong Chang, Korea, Republic of: IEEE, Feb. 2025, pp. 377–382. doi: 10.23919/ICACT63878.2025.10936757.

[23] A. Vishnu and S. Arora, “READS: Resource efficient attack detection system for drones,” *Multimed. Tools Appl.*, Oct. 2024, doi: 10.1007/s11042-024-20410-9.

[24] Y. Han, Z. Jia, S. He, Y. Zhang, and Q. Wu, “CNN+Transformer Based Anomaly Traffic Detection in UAV Networks for Emergency Rescue,” 2025, *arXiv*. doi: 10.48550/ARXIV.2503.20355.

[25] S. S. Alotaibi, A. Sayed, E. Samir Abd Elhameed, O. Alghushairy, M. Assiri, and S. Saadeldeen Ibrahim, “Enhancing Security in IoT-Assisted UAV Networks Using Adaptive Mongoose Optimization Algorithm With Deep Learning,” *IEEE Access*, vol. 12, pp. 63768–63776, 2024, doi: 10.1109/ACCESS.2024.3392618.

[26] I. A. Khan, D. Pi, S. Kamal, M. Alsuhaibani, and B. M. Alshammari, “Federated-Boosting: A Distributed and Dynamic Boosting-Powered Cyber-Attack Detection Scheme for Security and Privacy of Consumer IoT,” *IEEE Trans. Consum. Electron.*, vol. 71, no. 2, pp. 6340–6347, May 2025, doi: 10.1109/TCE.2024.3499942.

[27] A. A. Korba, A. Boualouache, and Y. Ghamri-Doudane, “Zero-X: A Blockchain-Enabled Open-Set Federated Learning Framework for Zero-Day Attack Detection in IoV,” *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 12399–12414, Sep. 2024, doi: 10.1109/TVT.2024.3385916.

[28] I. A. Khan *et al.*, “Fed-Inforce-Fusion: A federated reinforcement-based fusion model for security and privacy protection of IoMT networks against cyber-attacks,” *Inf. Fusion*, vol. 101, p. 102002, Jan. 2024, doi: 10.1016/j.inffus.2023.102002.

[29] T. M. Tran, D. C. Bui, T. V. Nguyen, and K. Nguyen, “Transformer-Based Spatio-Temporal Unsupervised Traffic Anomaly Detection in Aerial Videos,” *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 9, pp. 8292–8309, Sep. 2024, doi: 10.1109/TCSVT.2024.3376399.

[30] R. Majumder, G. Comert, D. Werth, A. Gale, M. Chowdhury, and M. S. Salek, “Graph-Powered Defense: Controller Area Network Intrusion Detection for Unmanned Aerial Vehicles,” 2024, *arXiv*. doi: 10.48550/ARXIV.2412.02539.

[31] S. Wei, Z. Fan, G. Chen, E. Blasch, Y. Chen, and K. Pham, “TADAD: Trust AI-based Decentralized Anomaly Detection for Urban Air Mobility Networks at Tactical Edges,” in *2024 Integrated Communications, Navigation and Surveillance Conference (ICNS)*, Herndon, VA, USA: IEEE, Apr. 2024, pp. 1–10. doi: 10.1109/ICNS60906.2024.10550825.

[32] M. Andreoni, W. T. Lunardi, G. Lawton, and S. Thakkar, “Enhancing Autonomous System Security and Resilience With Generative AI: A Comprehensive Survey,” *IEEE Access*, vol. 12, pp. 109470–109493, 2024, doi: 10.1109/ACCESS.2024.3439363.

[33] H. El Alami and D. B. Rawat, “DroneDefGANt: A Generative AI-Based Approach for Detecting UAS Attacks and Faults,” in *ICC 2024 - IEEE International Conference on Communications*, Denver, CO, USA: IEEE, Jun. 2024, pp. 1933–1938. doi: 10.1109/ICC51166.2024.10622524.

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[34] S. Silalahi, T. Ahmad, H. Studiawan, E. Anthi, and L. Williams, “Severity-Oriented Multiclass Drone Flight Logs Anomaly Detection,” *IEEE Access*, vol. 12, pp. 64252–64266, 2024, doi: 10.1109/ACCESS.2024.3396926.

[35] Z. Chen, I. Traoré, M. Mamun, and S. Saad, “Drone Anomaly Detection: Dataset and Unsupervised Machine Learning,” in *Foundations and Practice of Security*, vol. 15532, K. Adi, S. Bourdeau, C. Durand, V. Viet Triem Tong, A. Dulipovici, Y. Kermarrec, and J. Garcia-Alfaro, Eds., in Lecture Notes in Computer Science, vol. 15532. , Cham: Springer Nature Switzerland, 2025, pp. 186–201. doi: 10.1007/978-3-031-87499-4_12.

[36] S. Ullah *et al.*, “A New Intrusion Detection System for the Internet of Things via Deep Convolutional Neural Network and Feature Engineering,” *Sensors*, vol. 22, no. 10, p. 3607, May 2022, doi: 10.3390/s22103607.

[37] R. Alghamdi and M. Bellaiche, “An ensemble deep learning based IDS for IoT using Lambda architecture,” *Cybersecurity*, vol. 6, no. 1, p. 5, Mar. 2023, doi: 10.1186/s42400-022-00133-w.

[38] S. M. Kasongo and Y. Sun, “Performance Analysis of Intrusion Detection Systems Using a Feature Selection Method on the UNSW-NB15 Dataset,” *J. Big Data*, vol. 7, no. 1, p. 105, Dec. 2020, doi: 10.1186/s40537-020-00379-6.

[39] Y. N. Kunang, S. Nurmaini, D. Stiawan, and B. Y. Suprapto, “Attack classification of an intrusion detection system using deep learning and hyperparameter optimization,” *J. Inf. Secur. Appl.*, vol. 58, p. 102804, May 2021, doi: 10.1016/j.jisa.2021.102804.

[40] R. Zhao *et al.*, “A Novel Traffic Classifier With Attention Mechanism for Industrial Internet of Things,” *IEEE Trans. Ind. Inform.*, vol. 19, no. 11, pp. 10799–10810, Nov. 2023, doi: 10.1109/TII.2023.3241689.

[41] Y. Gorishniy, I. Rubachev, V. Khrulkov, and A. Babenko, “Revisiting deep learning models for tabular data,” in *Proceedings of the 35th International Conference on Neural Information Processing Systems*, in NIPS ’21. Red Hook, NY, USA: Curran Associates Inc., Dec. 2021, pp. 18932–18943.

[42] X. Huang, A. Khetan, M. Cvitkovic, and Z. Karnin, “TabTransformer: Tabular Data Modeling Using Contextual Embeddings,” Dec. 11, 2020, *arXiv*: arXiv:2012.06678. doi: 10.48550/arXiv.2012.06678.

[43] Q. Liu, W. Yang, C. Liang, L. Pang, and Z. Zou, “Tokenize features, enhancing tables: the FT-TABPFN model for tabular classification,” Jun. 11, 2024, *arXiv*: arXiv:2406.06891. doi: 10.48550/arXiv.2406.06891.

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